ECE 447

Lesson 40

Advanced Topics: OFDM, MIMO, CDMA



Schedule and Admin

- Schedule.
  - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
  - Lesson 41 - Course review
  - **Final Exam** – 0730-1120, Saturday, 12 Dec
- Admin
  - **HW9**. Due tonight (Lesson 40, 7 Dec) to Gradescope.
  - **Lab 8**. Due Lesson 41 (Wed, 9 Dec) to Gradescope.
  - **Extra Credit** (optional). Due Lesson 41 (Wed, 9 Dec).

What We Will Attempt to Cover Today

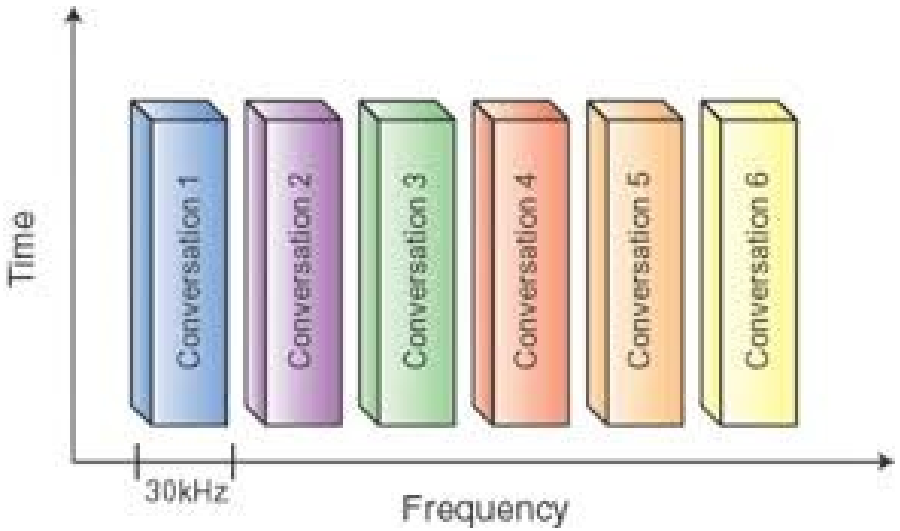
- **Multiplexing & multiple access** – FDM/A, TDM/A, CDM/A, SDM/A, PDM/A, OFDM/A
- **OFDM** – subcarriers, orthogonality, and why modern digital comm relies on it
- **Cyclic prefix** – what it is and why OFDM cannot do without it
- **Frequency-selective fading** – channel equalization, pilot signals, and CSI
- **MIMO** – spectral efficiency, and array/diversity/coding gains

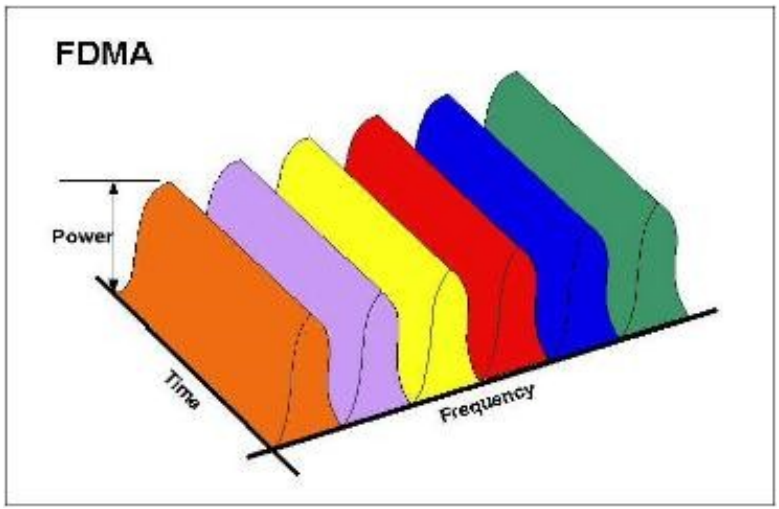
Multiplexing or Multiple Access?

- Both refer to how resources in a communication system are shared
- **Multiplexing**
  - Fixed or slowly changing allocation
  - Assigned a priori
- **Multiple access**
  - Dynamic allocation
  - Controller must be aware of each user's needs
    - How? Through control signals that run in background
    - Why? Enables efficient utilization of resources

FDM/A

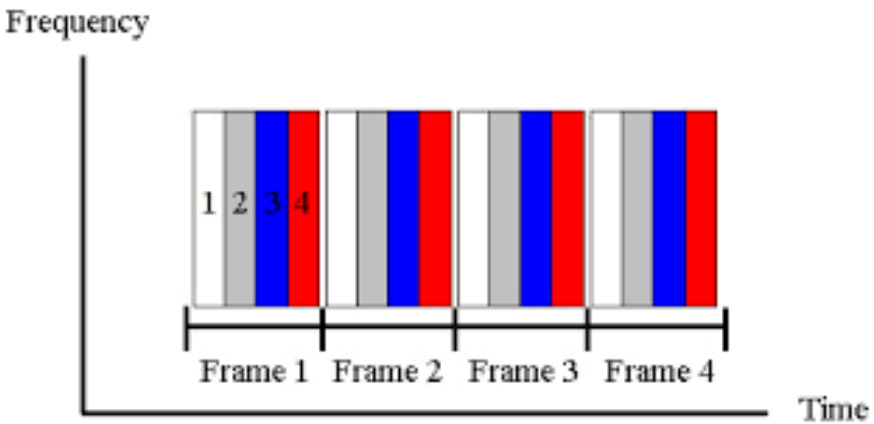
- Allowing multiple users to use the same channel by splitting up frequency channels
  - Separation done in the frequency domain

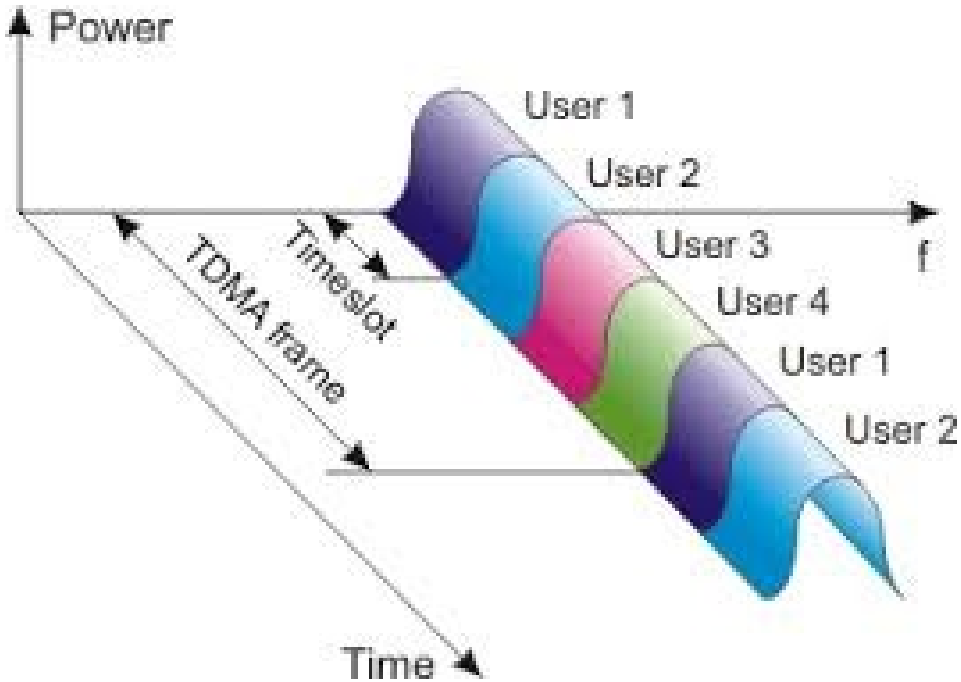




TDM/A

- Allowing multiple users to use the same channel by splitting up time slots
  - Separation done in the time domain





CDM/A

- Share communications resources by using special non-interfering/orthogonal codes that can be transmitted at the same time and on the same frequency

Data at Transmitter



Data at Receiver



CDM/A Example – Spread Spectrum

Direct-Sequence Spread Spectrum example:

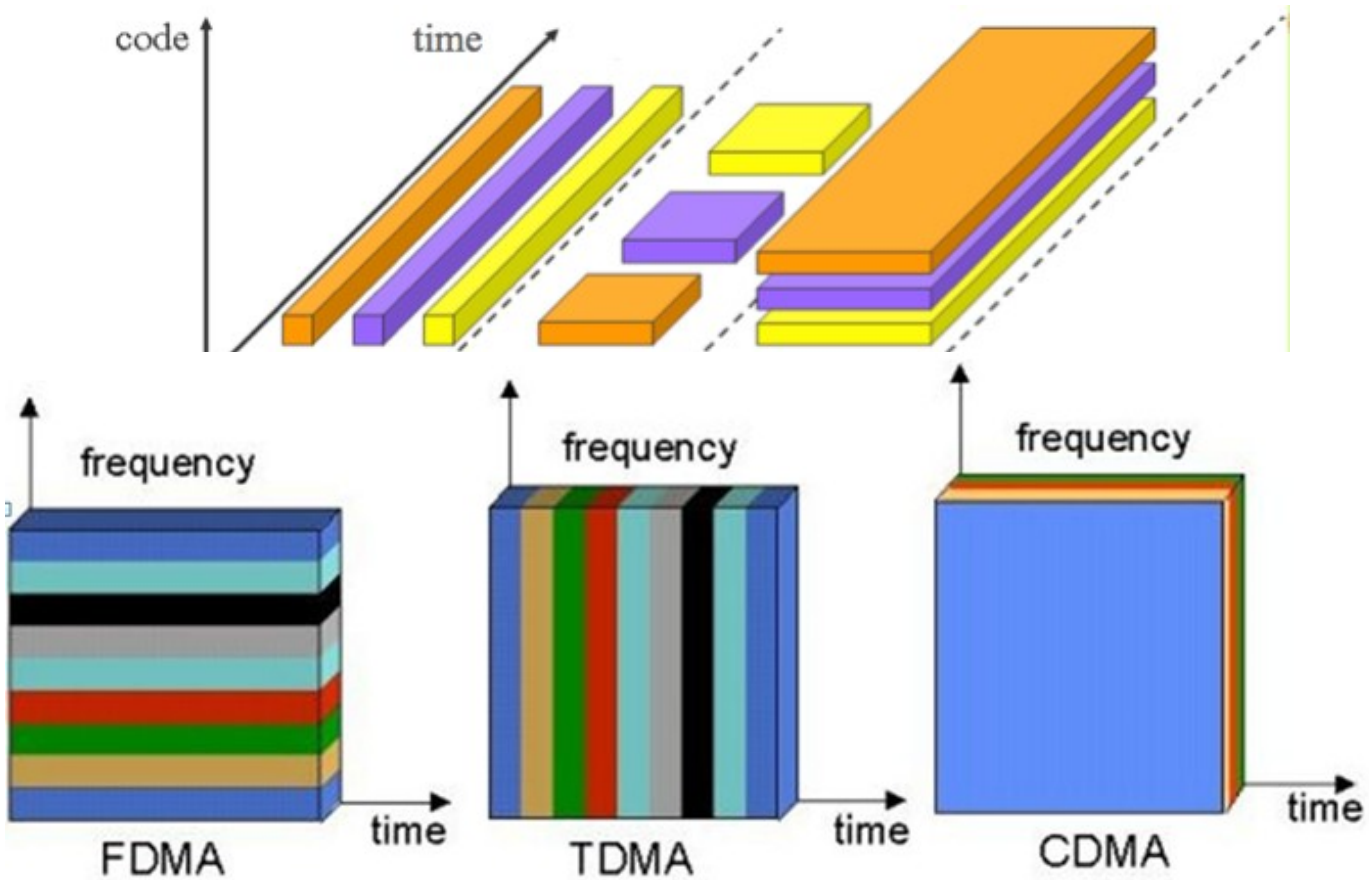
Bit to be transmitted:	`1`	(this bit lasts 10 μs)
Chipped bit (resampled)	`1111111111`	(each "chip" is 1 μs)
PRN	`1011010101`	
XOR result	`0100101010`	XOR: one + one = 0; one + zero = 1

The sequence of `0100101010` is transmitted to the receiver:

Bits received:	`0100101010`	
Receiver applies PRN	`1011010101`	PRN is exactly the same as above
XOR result	`1111111111`	

This string of ten "1"s is passed through a low pass filter which recreates the original 10 μs-long bit.

Comparing Methods

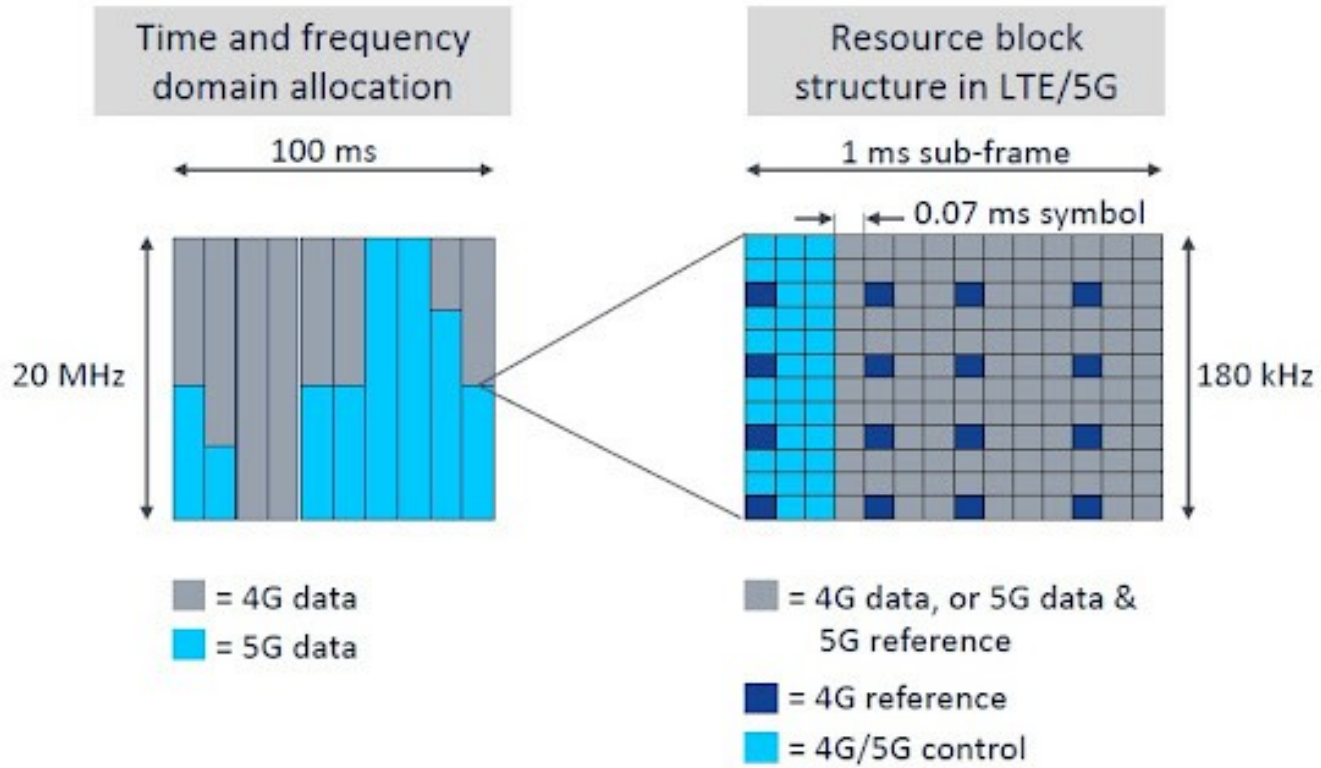


Example: 5G Resource Blocks

NOKIA

White paper
Nokia dynamic spectrum sharing for rapid 5G coverage rollout

Figure 3. 4G and 5G resource sharing in time and frequency domain



Via The 3G4G Blog - blog.3g4g.co.uk

SDM/A and PDM/A

- **SDM/A**
  - Use multiple antennas with different beams separated spatially
  - Prevalent in satellite and 4G LTE/5G cellular systems
- **PDM/A**
  - Antenna polarization determined by direction of electric field
  - Antennas with ideal orthogonal polarizations do not interfere with each other
  - Examples include satellite TV and cellular backhaul (microwave antennas)

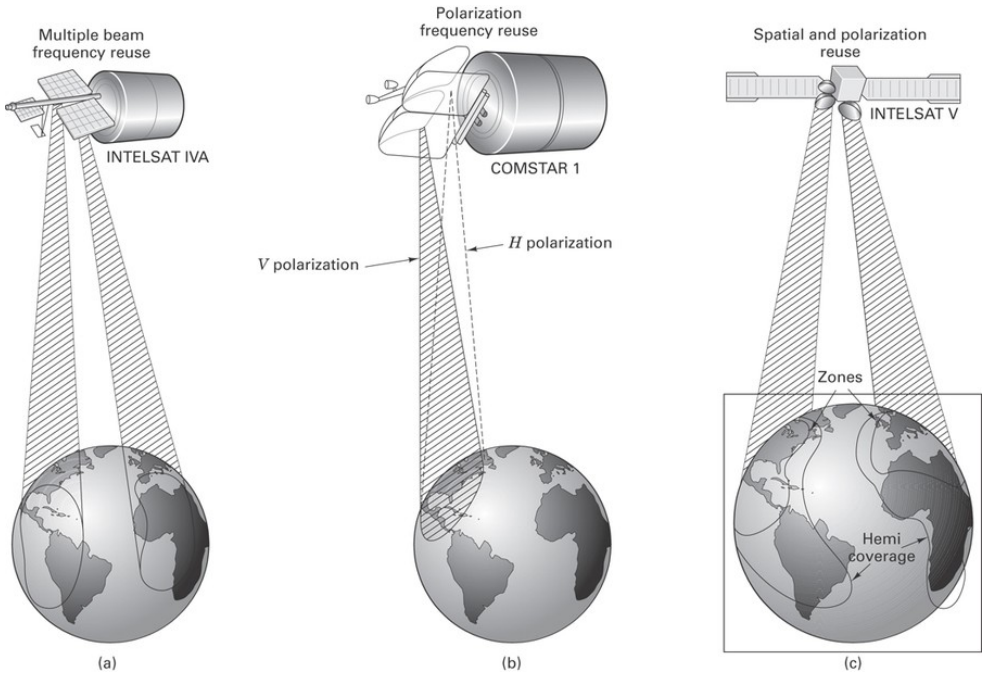
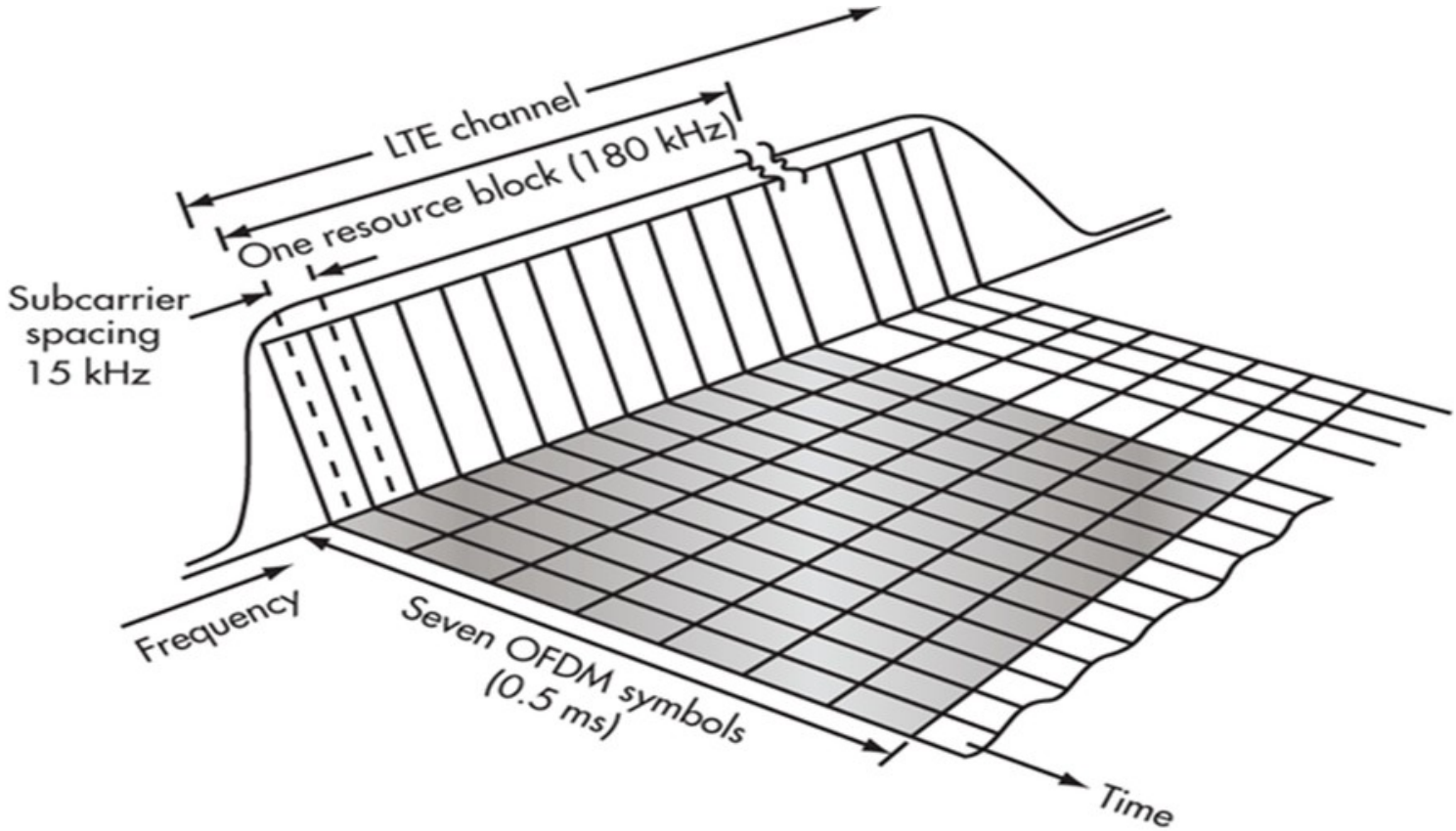


Figure 11.15 from Sklar

OFDM/A

- Share communications resources in both the frequency and time domains simultaneously, by transmitting on separate channels during specific time slots

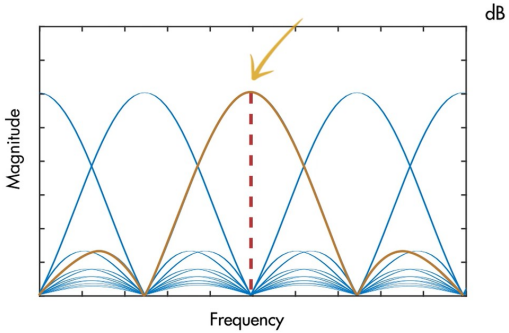


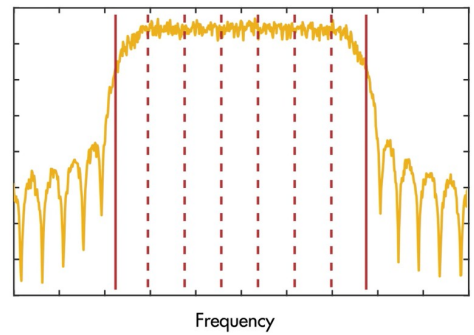
What is OFDM?

- A technique used to transform a single serial high-rate data stream into many parallel low-rate data streams
- Each QAM symbol from serial stream is assigned to a different frequency within the signal bandwidth known as a **subcarrier**
  - The OFDM symbol is the sum of all frequency-translated QAM symbols
- Each subcarrier peak lines up at the zero crossing of the others → this makes them **orthogonal**!



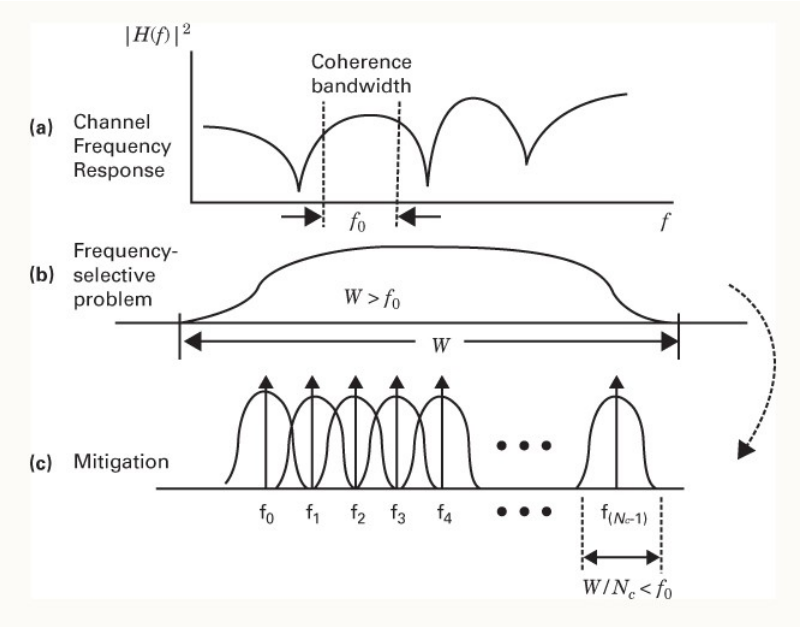
Figure 15.1 Transport analogy: One large delivery vehicle depicts ordinary FDM. Many small vehicles depict OFDM. (Charan Langton, www.complextoreal.com)





Why OFDM?

- Need **coherence bandwidth** of channel to be greater than signal bandwidth — the BW over which the channel's amplitude response is flat and phase response is linear (relatively)
  - High data rates → wideband signals → large coherence BW
  - Unlikely…why?
- Instead OFDM divides signal bandwidth into many small narrowband channels with BW less than the channel coherence BW
- Why do we care if these subcarriers are orthogonal or not?



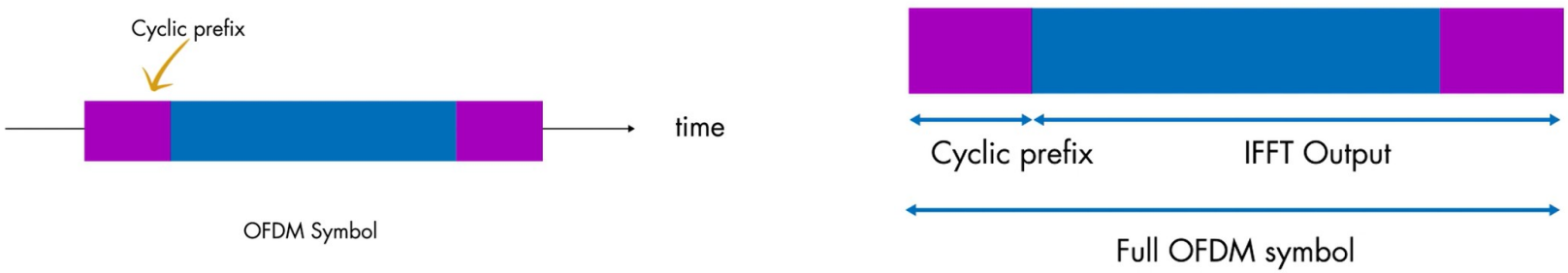
Bottom line

OFDM allows us to send spectrally-efficient, wideband (high data rate) signals across channels we normally would not be able to

How do we make and transmit OFDM symbols?

1. Move QAM symbols to subcarrier frequencies and sum together
2. Add **cyclic prefix**
   - Necessary for accurate channel equalization

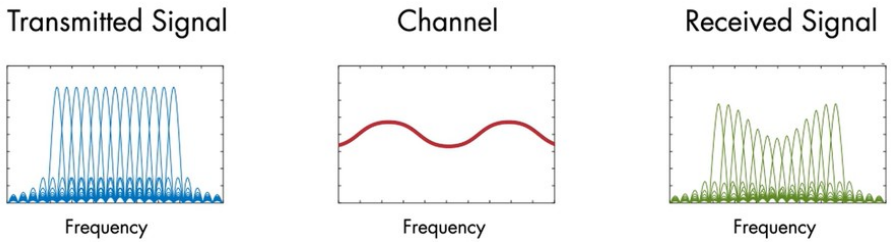
$$s(k) = \sum_{m=0}^{N-1} a_m e^{j2\pi(\frac{mk}{N})}$$



3. Then upconvert to desired transmission frequency, amplify to desired power, and transmit

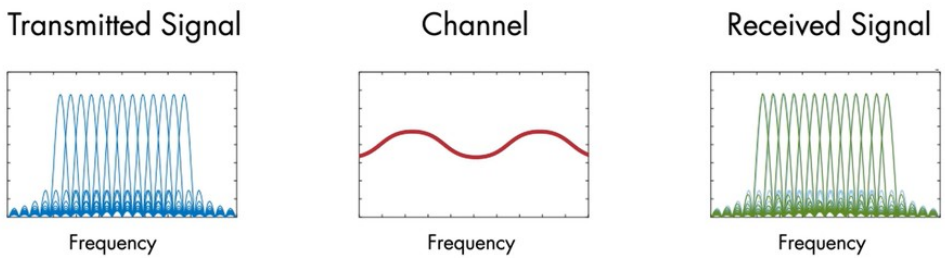
What about the channel distortion?

- Recall that the channel will affect the subcarrier amplitudes



- We use our knowledge of the channel, or the **channel model** $H[k]$, to perform **equalization** on the received OFDM symbols

Remember: in the time domain the received signal is the convolution of the transmitted signal and the channel's impulse response. What is convolution in the frequency domain?

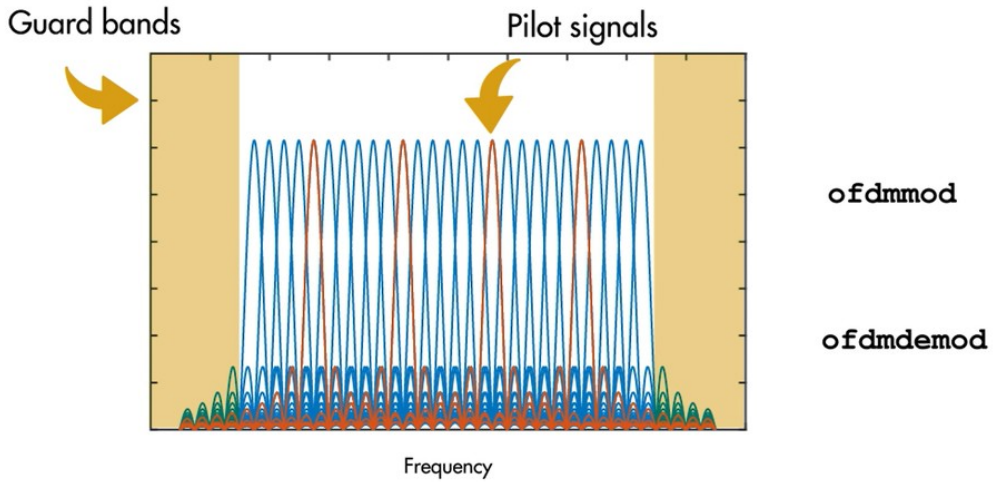


$$\hat{S}[k] = \frac{R[k]}{\hat{H}[k]}$$

How do we find out what the channel is?

- Evenly spaced subcarriers in the OFDM symbol are designated as **pilot signals**
- Known pilot signals are used to estimate the channel model $H[k]$ to get the **Channel State Information (CSI)**
  - Decisions on modulation order, rate allocation, power level, scheduling, and more based on CSI
- **Guardbands** separate OFDM symbols

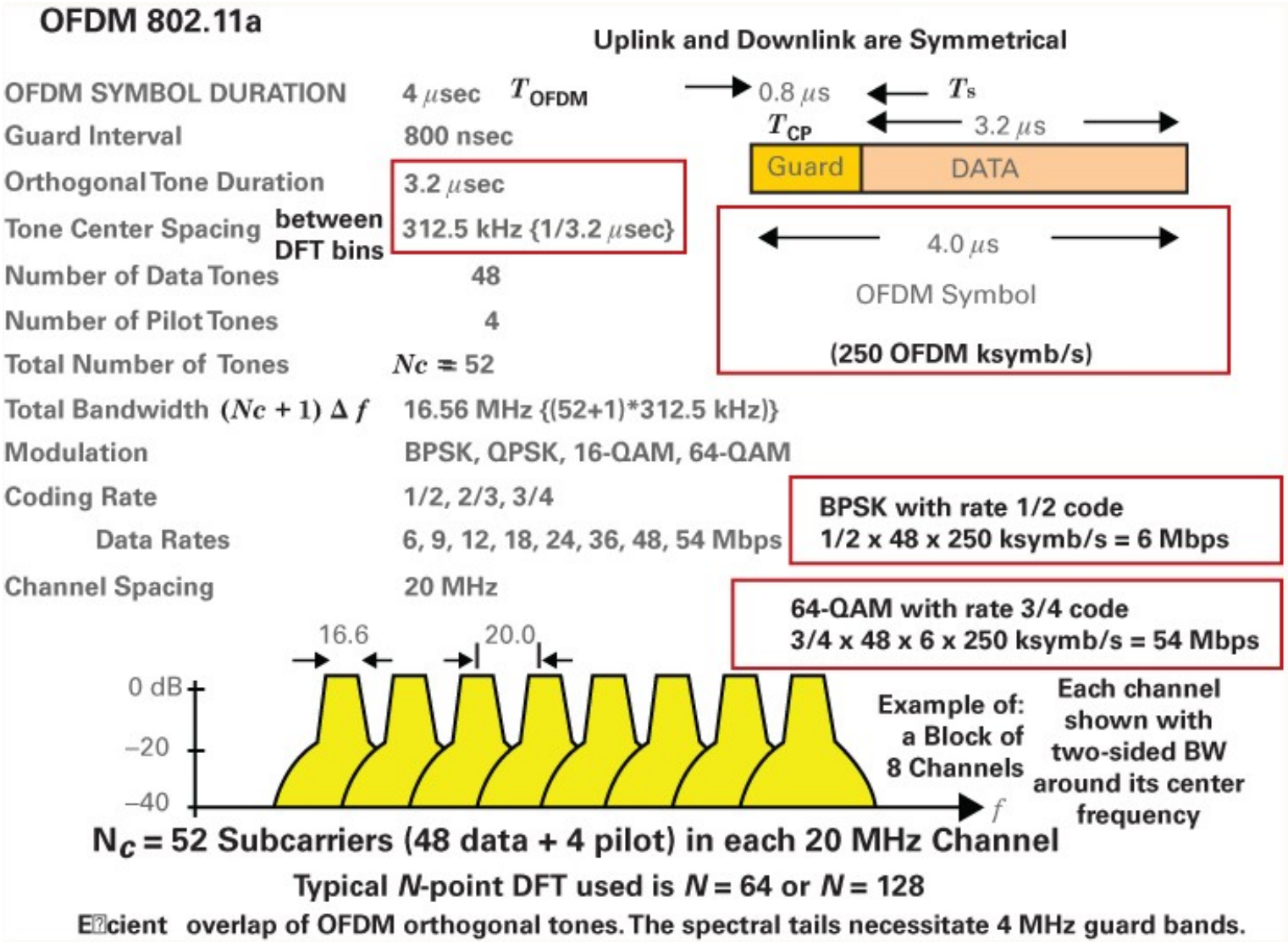
So . . . what happens if CSI is wrong?



How do we receive OFDM symbols?

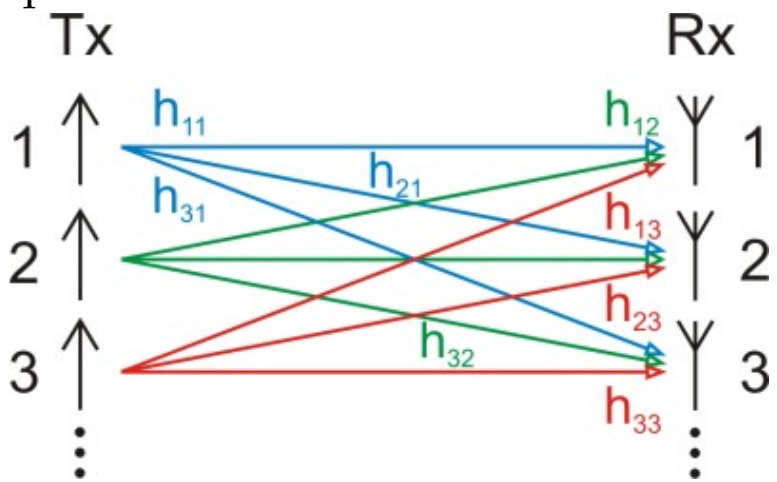
- Downconvert from carrier frequency
- Remove cyclic prefix
  - Doesn't affect OFDM symbol since cyclic prefix is redundant
- Use CSI to equalize received QAM symbols at their respective subcarrier frequencies
- Demodulate QAM symbols to bits

OFDM IN 802.11A (WIFI)



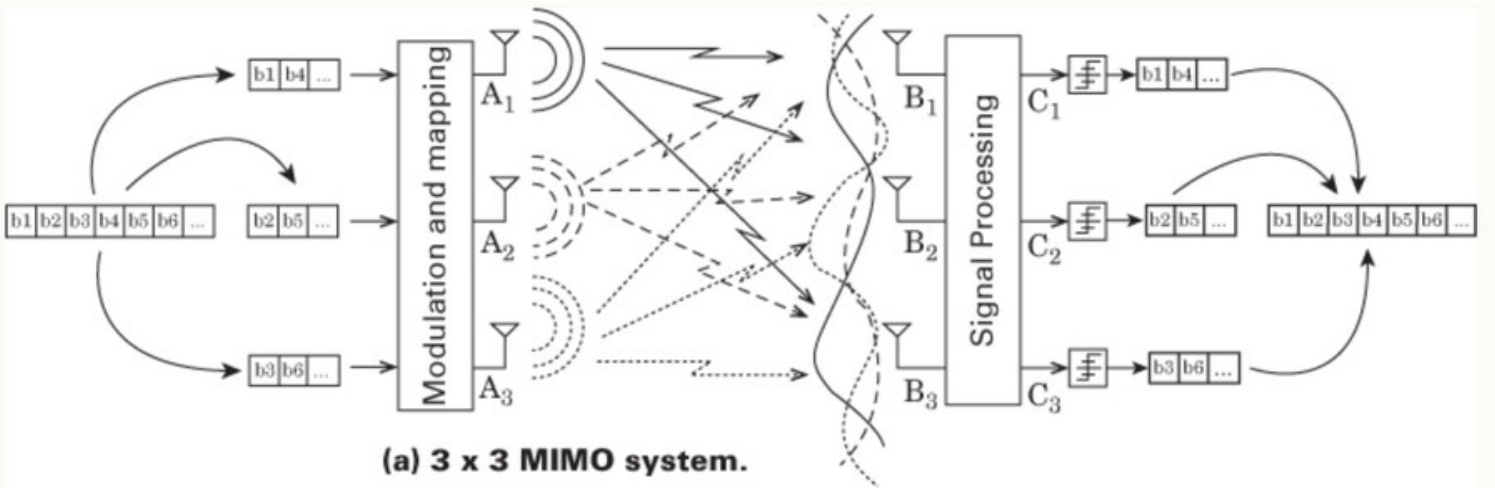
What is MIMO?

- MIMO systems employ multiple antennas at the Tx and Rx
- **Space-time signal processing** – includes spatial component where previously we only used time/freq
  - Improves BER and/or increases channel capacity (theoretical data rate) without requiring more power or additional bandwidth
  - Instead of "fixing" multipath environment – requires it
- Channel state information represented by matrix ***H***
  - Linear algebra! SVD! PCA!

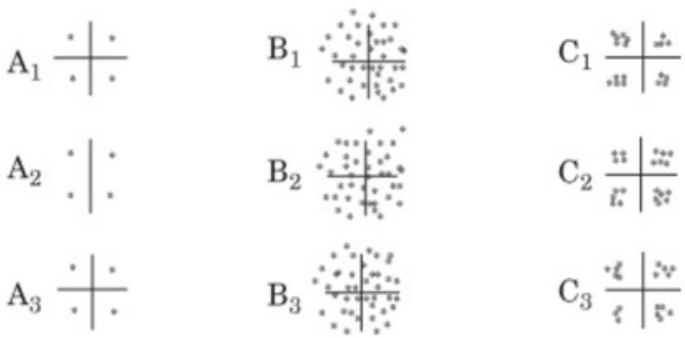


EXAMPLE: 3X3 MIMO

- Spread spectrum analogy: instead of different signals sent with different codes sharing the same BW at a cost of expanded BW, different signals are sent over different paths sharing the same BW at a cost of additional antennas



(a) 3 x 3 MIMO system.



(b) Signal constellations.

Importance of MIMO

- Foschini (Bell Labs) demonstrated spectral efficiency greater than 40 bits/s/Hz in 1996
  - Appeared to defy theoretical "Shannon limit"
  - For a conventional single antenna system, each bit/s/Hz requires doubling the constellation size starting with BPSK
  - Extreme spectral efficiency made possible using multiple antennas at the Tx and Rx
- Space-time coding provides coding gain, but not from using additional bits like in FEC
  - Uses additional antennas
  - Diversity gain improves robustness across varying channels

Other MIMO Concepts

- If Rx and Tx have CSI…
  - Tx able to optimally allocate power to channels with high SNRs
  - Called **water-filling algorithm**
- **Multi-User MIMO (MU-MIMO)**
  - More complicated – can't perform joint computations (channel matrix inversions)
  - Requires cross-layer design
  - PHY layer modulation and coding, Link layer multiple-access channel
  - WiFi 802.11ac standard uses beamforming MU-MIMO in downlink direction

